F78AA1 – Actuarial and Financial Mathematics A

December 2011 Examination – General Feedback

This document provides some general feedback on the December 2011 exam. The paper was generally attempted well, including some of the more challenging 'unseen' questions. However, these questions (Q3c)ii), Q5 and Q6) did tend to clearly separate stronger scripts from weaker ones.

A significant number of scripts were penalised heavily for giving a wrong answer with insufficient working to show how that answer was reached. Showing interim working clearly is *very* important: without it, there is no evidence to the examiner that you understand the calculation procedure even if, in reality, the only error you have made is a slip up with your calculator.

Question 1

- There were a wide range of marks for this question, but with few scoring full marks.
- The main issue was with accurately understanding the definitions and converting them into actuarial notation.
- In particular, many scripts calculated *v* (the discount *factor*) in part ii) rather than *d* (the discount *rate*).
- Many scripts also calculated the value of the annuity in part iii) annually and/or in arrears, rather than half yearly in advance – read the question carefully!

Question 2

- This question was generally answered very well, with the majority of scripts scoring close to full marks.
- The main issues in those scripts which scored less well were in relation to the calculation in part a). In particular:
    - Only accumulating the first and second years' cashflows to the end of the first and second year respectively (rather than right up to the end of the third year);
    - Accumulating each cashflow individually, rather than applying notation for annuities. Where no annuity functions were used, very little working tended to be shown except for a final answer, which was often incorrect. As such, little credit could be given.

Question 3

- Part a): Scripts tended to either receive full marks or zero marks here – this proof was taken straight from our notes, and a significant (and disappointing) number of students failed to produce it;
- Part b) – very well answered, with most scripts receiving full marks;
- Part c)i) – likewise, very well answered;
- Part c)ii) – Again, scripts tended to score either full marks or zero marks here. The trick in this previously unseen loan alteration is to recognise that we can essentially treat it in the same way as any other. i.e. First calculate the capital outstanding at the time of the alteration (part c)i)), then equate this to the value (at the time of the alteration) of the future repayments (here, in the form of a compound increasing annuity).

Question 4

- Part a) was a standard question on fixed interest securities and was generally answered well.
- The main issues were
    - Not first checking if capital gains tax was/was not payable using the formula from the notes;
    - Some who did use this formula incorrectly applied it on the basis of a time unit of years, rather than half-years: the formula needs to be applied in the same time unit as the cashflows themselves.
- Part b) tested the relationship between the real yield, monetary yield and inflation rate using the formula from the final section of our notes.

Question 5

- Part a) was generally answered well.
- A good number of scripts made a very good attempt at part b), assuming that the Discounted Payback Period was at the time of the 7^{th} cashflow returned.
- The main issues were
    - Lack of awareness of how to value a simple increasing annuity;
    - Not checking that the initial investment had *not* been recovered at the time of the 6^{th} cashflow.
- There was an ambiguity in the wording to Part b) which affected a minority of scripts – those who were misled but clearly showed an awareness of how to value a simple increasing annuity, as well as how to go about finding the discounted payback were rewarded.

Question 6

- Part a) was answered well.
- An reasonable number of scripts made a strong attempt at part b)i) - a basic MWRR calculation but involving interpolation, which many applied well.
- For part b)ii), those who noticed that they had to formulate an equation of value for the TWRR but solving for a different unknown (the fund value) scored well.